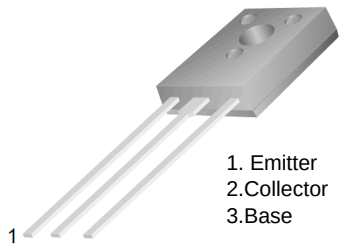
# NPN Epitaxial Silicon Transistor

# BD135, 137, 139




**TO-126-3LD**
**CASE 340AS**

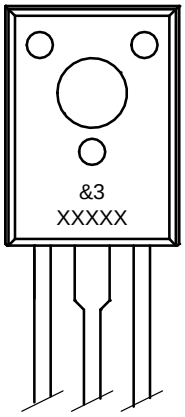
### Features

- Complement to BD136, BD138 and BD140 Respectively
- These are Pb-Free Devices

### Applications

- Medium Power Linear and Switching

### MARKING DIAGRAM




&3 = 3-Digit Date Code
XXXXX = Specific Device Code

### ABSOLUTE MAXIMUM RATINGS ($T_C$ = 25 °C unless otherwise noted)

| Symbol | Parameter | | Ratings | Units |
|---|---|---|---|---|
| $V_{CBO}$ | Collector-Base Voltage | BD135 | 45 | V |
| | | BD137 | 60 | |
| | | BD139 | 80 | |
| $V_{CEO}$ | Collector-Emitter Voltage | BD135 | 45 | V |
| | | BD137 | 60 | |
| | | BD139 | 80 | |
| $V_{EBO}$ | Emitter-Base Voltage | | 5 | V |
| $I_C$ | Collector Current (DC) | | 1.5 | A |
| $I_{CP}$ | Collector Current (Pulse) | | 3.0 | A |
| $I_B$ | Base Current | | 0.5 | A |
| $P_C$ | Device Dissipation | $T_C$ = 25 °C | 12.5 | W |
| | | $T_A$ = 25 °C | 1.25 | W |
| $T_J$ | Junction Temperature | | 150 | °C |
| $T_{STG}$ | Storage Temperature | | −55 to +150 | °C |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

### THERMAL CHARACTERISTICS

| Symbol | Parameter | Max | Units |
|---|---|---|---|
| $R_{\theta JC}$ | Thermal Resistance, Junction-to-Case | 10 | °C/W |
| $R_{\theta JA}$ | Thermal Resistance, Junction-to-Ambient | 100 | °C/W |

### ORDERING INFORMATION

See detailed ordering, marking and shipping information on page 4 of this data sheet.

NOTE: Some of the devices on this data sheet have been **DISCONTINUED**. Please refer to the table on page 4.

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**ELECTRICAL CHARACTERISTICS** ($T_C$ = 25 °C unless otherwise noted)

| Symbol | Parameter | | Test Condition | Min | Typ | Max | Unit |
|---|---|---|---|---|---|---|---|
| $V_{CEO}$(sus) | Collector-Emitter Sustaining Voltage | BD135 | $I_C$ = 30 mA, $I_B$ = 0 | 45 | | | V |
| | | BD137 | | 60 | | | |
| | | BD139 | | 80 | | | |
| $I_{CBO}$ | Collector Cut-off Current | | $V_{CB}$ = 30 V, $I_E$ = 0 | – | | 0.1 | μA |
| $I_{EBO}$ | Emitter Cut-off Current | | $V_{EB}$ = 5 V, $I_C$ = 0 | – | | 10 | μA |
| $h_{FE1}$ | DC Current Gain | | $V_{CE}$ = 2 V, $I_C$ = 5 mA | 25 | | | |
| $h_{FE2}$ | | | $V_{CE}$ = 2 V, $I_C$ = 0.5 A | 25 | | | |
| $h_{FE3}$ | | | $V_{CE}$ = 2 V, $I_C$ = 150 mA | 40 | | 250 | |
| $V_{CE}$(sat) | Collector-Emitter Saturation Voltage | | $I_C$ = 500 mA, $I_B$ = 50 mA | | | 0.5 | V |
| $V_{BE}$(on) | Base-Emitter On Voltage | | $V_{CE}$ = 2 V, $I_C$ = 0.5 A | | | 1 | V |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.



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## TYPICAL PERFORMANCE CHARACTERISTICS

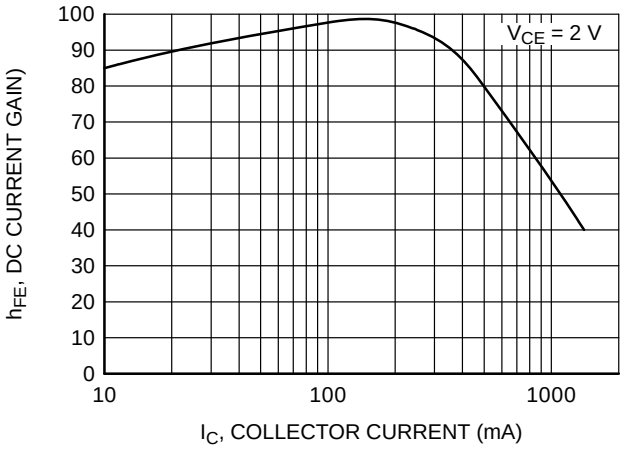



**Figure 1. DC Current Gain**

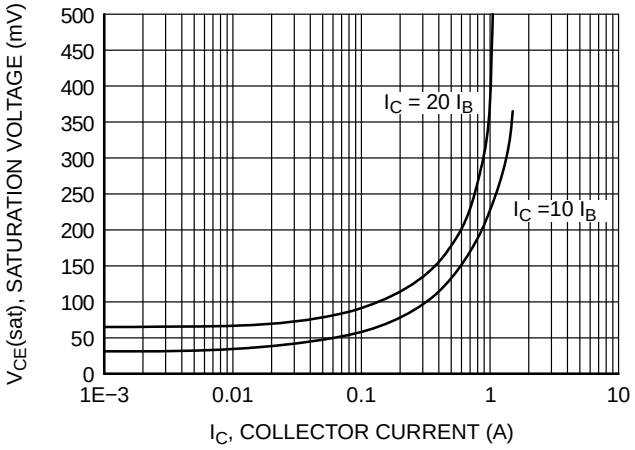



**Figure 2. Collector-Emitter Saturation Voltage**

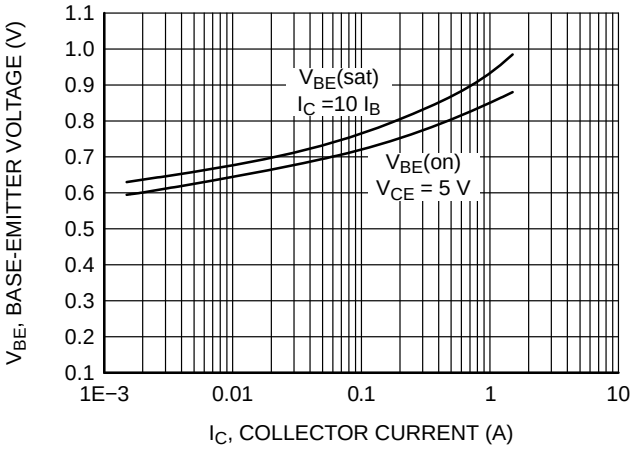



**Figure 3. Base-Emitter Voltage**

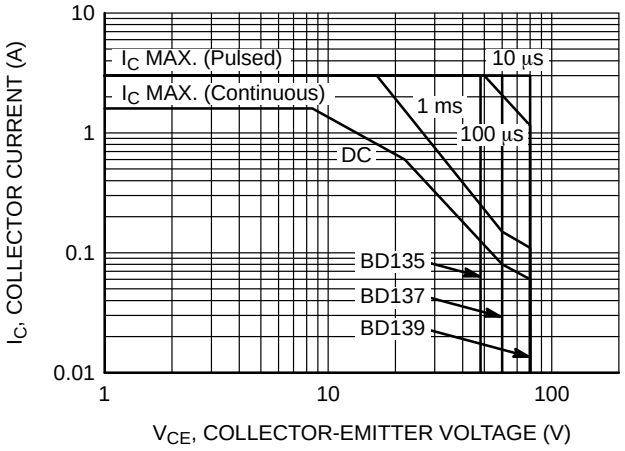



**Figure 4. Safe Operating Area**

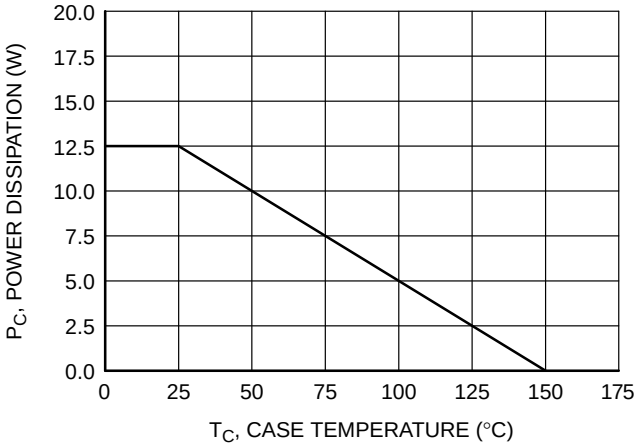



**Figure 5. Power Derating**



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## ORDERING INFORMATION

| Part Number | Marking | Package | Packing Method |
|---|---|---|---|
| BD13516STU | BD135-16 | TO-126-3LD (Pb-Free) | Rail |
| BD13716STU | BD137-16 | | |
| BD13916STU | BD139-16 | | |

**DISCONTINUED** (Note 1)

| | | | |
|---|---|---|---|
| BD13516S | BD135-16 | TO-126-3LD (Pb-Free) | Rail |
| BD1356STU | BD135-6 | | |
| BD13510STU | BD135-10 | | |
| BD13710STU | BD137-10 | | |
| BD13716S | BD137-16 | | Bulk |
| BD13910S | BD139-10 | | |
| BD13916S | BD139-16 | | |
| BD1396STU | BD139-6 | | Rail |
| BD13910STU | BD139-10 | | |

1. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.



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## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 3 | Document converted to **onsemi** format.<br>Added Thermal Characteristics, removed $h_{FE}$ Classification.<br>BD13516S, BD1356STU, BD13510STU, BD13710STU, BD13716S, BD13910S, BD13916S, BD1396STU, BD13910STU OPNs marked as Discontinued. | 4/28/2026 |

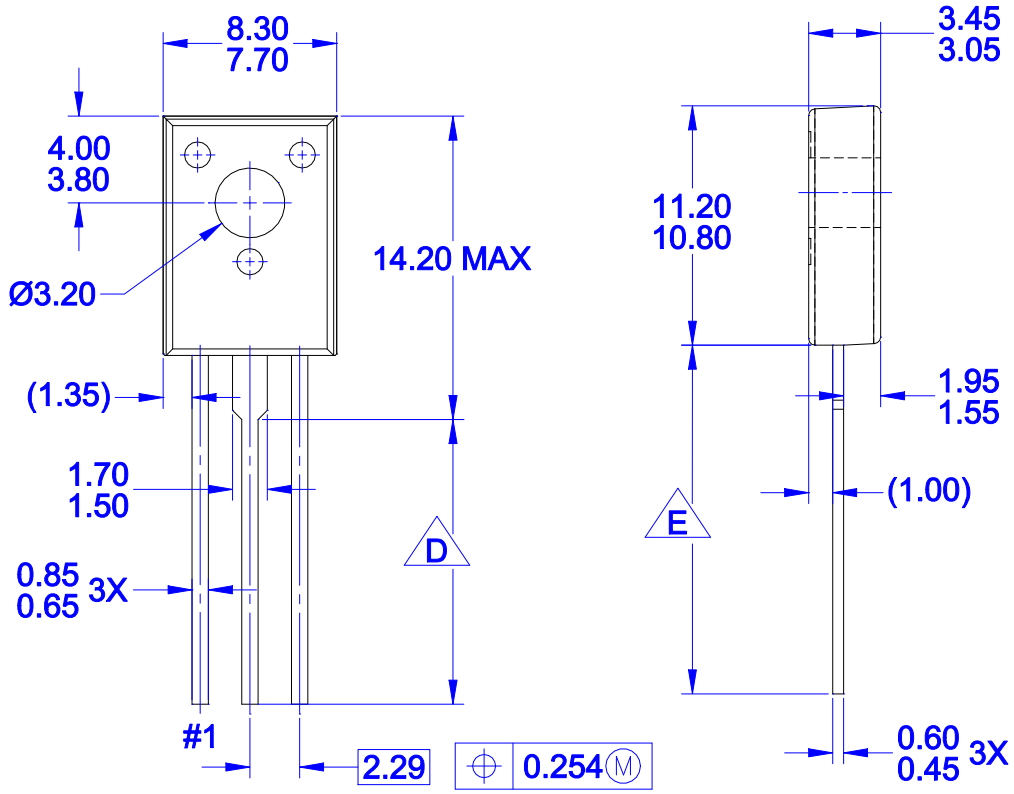
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.



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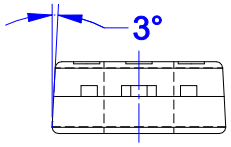
**TO−126−3LD**
CASE 340AS
ISSUE O

DATE 30 SEP 2016




TOP VIEW

SIDE VIEW




FRONT VIEW

| PRODUCTION CODE | TERMINAL LENGTH "D" | TERMINAL LENGTH "E" |
|---|---|---|
| TSSTU | 3.45 - 4.05 | 6.45 - 7.45 |
| TSTU | 2.36 - 2.96 | 5.36 - 6.36 |
| NONE (STD LENGTH) | 12.76 - 13.36 | 15.76 - 16.76 |

NOTES:

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B. ALL DIMENSIONS ARE IN MILLIMETERS
C. DIMENSIONS ARE EXCLUSIVE OF BURRS, MOLD FLASH, AND TIE BAR PROTRUSIONS

D FOR TERMINAL LENGTH "D", REFER TO TABLE

E FOR TERMINAL LENGTH "E", REFER TO TABLE

| | | | |
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